

Candidate biomarkers for precision monitoring of post-exercise immunity: a scoping review of conventional and omics evidence

Protocol version: 2.0 — 2 October 2026

Status: Pre-registration working protocol; formal searches, screening and synthesis have not begun.

Target journal: *Frontiers in Immunology*, under the applicable Systematic Review article category, which includes scoping reviews; specialty section to be selected.

Registration: OSF draft prepared locally; not submitted; no registration identifier.

Team: Access to all five databases and a second human reviewer are confirmed. Names, adjudicator, search peer reviewer, funding and competing-interest declarations remain to be supplied.

This is the canonical English protocol. The Chinese protocol is an aligned implementation version. The rendered English source and PDF are generated from this complete document; they are not a shortened independent protocol. Operational appendices in 03_search–06_synthesis implement the decisions below. This version follows public exploratory reconnaissance and source verification. The protocol date is not a completed formal search cutoff. No search yield, eligibility decision, included-study count or review result is claimed here.

A. Background and rationale

Exercise can alter circulating immune-cell counts, immune mediators, mucosal measurements and immune-cell function over different time scales. The historical “open window” hypothesis has linked some post-exercise changes to temporary immune suppression and infection susceptibility. These interpretations remain contested: a circulating cell-count reduction does not establish a loss of whole-body immune function, and immune responses do not by themselves establish subsequent infection risk. Redistribution, plasma-volume changes, circadian variation and sampling methods can affect interpretation [1,2]. This review treats the open window as a historical hypothesis and an author-reported interpretation, rather than an established phase or an eligibility requirement.

Conventional measures include leukocyte subsets, immune-cell effector assays, immunoglobulins, salivary secretory IgA, cytokines and complement. Transcriptomics, proteomics, metabolomics, lipidomics, non-coding RNA and epigenetic measurements provide additional candidates. Higher measurement dimensionality does not alone confer immune specificity or individual predictive value. A group-average response, an in-silico pathway annotation, an association with symptoms and a validated individual prediction address different questions. Candidate usefulness also depends on the exercise exposure, tissue or fluid, analytical method, sampling interval and intended monitoring use.

The intended contribution is an evidence map of **candidate biomarkers for precision monitoring**, emphasizing individual baselines, within-person variation, repeatability, time-resolved interpretation, functional or clinical links, and independent validation. This is not a catalogue that labels every statistically significant omics feature a precise biomarker. Analytical and clinical validation must be interpreted for a specified context of use [3].

Existing reviews address exercise-associated non-coding RNAs, B-cell-related immunity, open-window interpretations in older adults and exercise immunometabolomics [4–7]. Public reconnaissance has identified substantial adjacent literature. We therefore make no “first review” or “no competing review” claim. The proposed distinction is the joint mapping of conventional and omics evidence against explicitly defined post-exercise timing and separate precision-monitoring domains. That distinction will be rechecked against published and registered reviews before registration and at the final search update.

A scoping review is appropriate because the primary questions concern the distribution, measurement, interpretation and validation of heterogeneous evidence, rather than one intervention effect

or diagnostic threshold. We will follow JBI scoping-review methods [8], report with PRISMA-ScR [9], and report searches with PRISMA-S [10]. Prespecified, design-appropriate critical appraisal will support interpretation. We will not conduct a meta-analysis or turn appraisal into a clinical ranking.

B. Objective and review questions

The objective is to identify and map human evidence on immune-related candidate biomarkers measured after exercise and determine which aspects of precision monitoring have actually been investigated.

1. **Measurements and context:** Which biomarkers, analytical platforms, immune compartments and sample matrices have been studied, in which populations and exercise settings?
2. **Timing and interpretation:** Which response and recovery intervals are covered, and how are cell composition, hydration, sampling and other relevant confounders addressed?
3. **Individualization:** What evidence exists on personal baselines, within-person variability, repeatability, response heterogeneity, subgroup interpretation or individualized models, and how has it been validated?
4. **Function, outcomes and use:** Which functions or clinical outcomes were merely co-measured, which were associated with biomarkers, and which monitoring or prediction uses underwent independent validation? What development requirements remain?

Positive, null and inconsistent findings are all relevant. The questions do not presuppose that exercise is immunosuppressive or that any candidate is ready to predict infection.

C. Framework and operational definitions

Eligibility uses Population–Concept–Context (PCC). “Post-exercise” begins at cessation of an identifiable bout or competition. Exact times and their reported origin are retained. During-exercise samples can be ancillary observations in an eligible study but cannot alone establish post-exercise eligibility.

For this project, **precision monitoring** means interpreting an immune-related measurement for a defined person or subgroup, exercise context, sampling time and intended monitoring use. Seven domains are recorded separately:

Domain	Evidence to chart
Analytical reliability	Assay performance, limits, quality control, reproducibility and pre-analytical handling
Temporal validity	Exact exercise-to-sample intervals, longitudinal coverage and how response or recovery is defined
Immune specificity	Direct immune measurement, cellular origin, composition adjustment and relevant alternative explanations
Individualization	Personal baselines, within-person noise, repeated-bout reproducibility, response heterogeneity, subgroup or individual models
Functional or clinical linkage	Direct immune-function measurements, tested associations and explicitly defined clinical outcomes
Independent validation and use	Technical confirmation versus independent participants/cohorts, model validation and tested decision use

Domain	Evidence to chart
Feasibility	Sampling burden, accessibility, cost, turnaround and field implementation, when reported

These are project operational domains, not a validated scale, regulatory designation or ordinal development ladder. There will be no combined “precision score.” Omics measurement, machine learning or a group mean difference alone does not establish personalized usefulness. Individualization or clinical validation is **not required for inclusion**; absence of such reporting is itself a mapping finding.

A **response** is an observed post-exercise measurement or contrast. A **recovery trajectory** requires longitudinal observations and an explicit endpoint; a single early sample or a nonsignificant comparison with baseline does not establish recovery. Cell-count normalization, biomarker normalization, immune-function recovery and clinical recovery remain distinct.

Immune linkage requires a direct immune-cell/count/subset/effector measure; a recognized immune mediator such as an immunoglobulin, cytokine or complement component; omics measured in immune cells; or biofluid omics with an immune objective or analysis stated in the original aims/methods and identifiable immune candidates. Original-study preregistration is not required. Mere post-hoc immune enrichment in an otherwise general muscle or metabolic study is insufficient for the main map. Cortisol, creatine kinase, lactate or heart-rate variability alone do not qualify, but may be charted as covariates in otherwise eligible studies.

Direct functional assays are distinguished from computational annotation. Co-measurement of a function or clinical outcome does not establish a biomarker–outcome association. Self-reported symptoms, clinician-defined syndromes and pathogen-confirmed infection are separate outcome classes. Association, temporal prediction, calibration, discrimination and clinical utility are recorded separately. We use “prediction” for future outcome prediction here, without assuming the treatment-response meaning of a predictive biomarker in other taxonomies.

Missing information is coded **NR** (not reported), **NA** (not applicable) or **UNCLEAR** (information present but insufficient or ambiguous after checking). NR is not evidence of biological absence or methodological failure.

D. Eligibility criteria

D.1 Population

Eligible participants are humans of any age, sex or training status who are generally healthy at baseline and do not have an active infection. Athletes, recreational exercisers and untrained participants are eligible; age and training strata are charted separately. Extractable healthy strata within mixed cohorts are eligible. Cohorts with known clinical/metabolic disease or disease-treatment exercise programmes are excluded from the main map. Ambiguous metabolic-health descriptions, including insulin resistance without a clear clinical classification, require explicit adjudication using the reported diagnosis and separability of strata, rather than an automatic healthy designation.

D.2 Exercise context and evidence streams

Core A — acute challenge: An identifiable bout or competition of endurance, resistance, interval/sprint or mixed exercise, with at least one post-cessation immune-related sample and either a pre-bout baseline or a suitable non-exercise/comparator condition. Immediate samples are eligible. There is no minimum 30-minute follow-up and no maximum 72-hour interval. During-only

studies are ineligible. Repeated acute challenges before and after a training programme qualify for A; training moderation is extracted separately. Immediate-only studies contribute response evidence, not a complete recovery trajectory.

Support B — repeated post-exercise monitoring: The same immune-related marker is measured in the same individuals after at least two identifiable bouts or competitions, with a known or reliably recoverable exercise-to-sampling relation and a baseline or appropriate comparator enabling interpretation. B is presented separately. A cohort may meet both A and B; linked cohort identifiers prevent duplicate overall counting. General weekly or seasonal resting surveillance with unknown last-bout timing, training pre/post snapshots with unknown exercise-to-sampling intervals, and habitual-exercise cross-sectional comparisons are background only. A long observation period cannot substitute for an identifiable post-exercise time relation.

Studies involving supplementation, diet or other co-interventions qualify only where an exercise contrast is interpretable separately, for example through an appropriate control/placebo arm or factorial contrast. Otherwise the inseparable intervention is ineligible. Prior training, co-interventions and recovery arrangements are extracted.

D.3 Concept, design and publication status

Eligible studies meet the immune-link definition in C and report original human measurements. There is no requirement for the words “precision,” “biomarker,” “open window” or “recovery,” for a particular direction or significance level, or for a functional/infection endpoint. Designs may include randomized or nonrandomized experiments, crossover studies, before–after studies and observational designs that satisfy the timing and comparator rules. No arbitrary minimum sample size is imposed; small designs are clearly identified and appraised appropriately.

The primary A/B map includes peer-reviewed original reports. Reviews support background and citation chasing. Animal-only or in-vitro-only studies, protocols, editorials, conference abstracts without a full original report and pure methods papers without eligible human exercise observations are excluded from the primary map; potentially relevant abstracts are tracked for full publications.

A bounded supplementary search of bioRxiv and medRxiv will identify publicly available preprints meeting the same substantive PCC criteria. They enter a separately labelled frontier register, with separate flow counts and denominators, and are not pooled with peer-reviewed evidence. Publication status is checked at the final update, and preprint/published versions are linked and merged appropriately. Clearly labelled public preprints may be cited under current Frontiers guidance [14]; this does not imply access to unpublished private results.

D.4 Date, language and retrieval

Searches run from database inception to their actual execution dates. No search language restriction is applied. English and Chinese full texts are read directly; other languages use translation with independent human checking of eligibility and key data. Unresolved language or eligibility information remains awaiting classification rather than being automatically scientifically excluded.

Institutional access and lawful public copies will be used for full texts. A sought but unobtained report is “not retrieved,” not a scientific full-text exclusion. This protocol does not plan author outreach; no contact is sent without separate authorization. Retrieval attempts and unresolved reports are logged.

E. Information sources and search strategy

The five planned databases are PubMed, Web of Science Core Collection, Scopus, Embase and SPORTDiscus. PubMed is reported as PubMed rather than implying a separate MEDLINE-

platform search. Embase platform choice is unresolved: draft Embase.com and Ovid translations are supplied for confirmation; execution of one platform is not claimed as two databases. The SPORTDiscus draft uses EBSCOhost. Exact collections, platforms, interfaces and coverage will be recorded.

Two complementary routes are planned: **E AND I**, where E represents exercise and I immune concepts; and **E AND O**, where O represents omics. Their union is deduplicated. Controlled vocabulary and free-text fields are adapted to each platform. The immune route covers immune cells, mediators, function and relevant mucosal/infection terms. The omics route covers transcriptomics, RNA/non-coding RNA, proteomics, metabolomics, lipidomics, epigenetics and multi-omics. The exercise block includes relevant training, competition and sport terminology. “Precision,” “biomarker,” “open window” and “recovery” are not mandatory search concepts.

Full working strategies and a search-log template are in 03_search. They are **drafts, not platform-validated searches**. A search specialist or experienced independent search reviewer will use PRESS [11]; resulting revisions and resolution of comments will be retained. Known eligible and boundary examples will test strategy behavior. Retrieval of a few selected positive seeds is a debugging check, not an estimate of overall recall. A seed not retrieved will be investigated; strategy changes must retain a conceptual rationale rather than simply fitting known papers.

Backward references and forward citations of included reports and relevant neighboring reviews will be checked. Registered/ongoing reviews and adjacent recent reviews will inform the novelty assessment. Supplemental preprint searches will record sites, exact queries, dates, filters, results examined and stopping rules. Citation chasing has a defined sweep and documented closure/update; it is not silently unlimited browsing.

For every run, preserve the exact query, database/platform, run date and time, limits, hits, export counts, file format, immutable raw export and checksum. Record search amendments, supplementary routes and deduplication decisions under PRISMA-S. The target is to update searches within 30 days before submission; this is a project target, not a journal rule. New exports do not overwrite earlier exports.

Records are deduplicated using identifiers followed by normalized title/author/year checks and human confirmation of uncertain matches. Distinct reports of the same study or cohort remain linked, rather than being discarded as bibliographic duplicates. Study, report, cohort and preprint-version relationships are preserved. No formal yields are available at protocol preparation.

F. Selection and calibration

Two human reviewers independently screen **all** titles/abstracts and **all** sought full texts. At title/abstract stage, missing timing or methodological detail normally proceeds to full text rather than being guessed. AI does not make automatic exclusions or replace either reviewer.

Before formal screening, a saved random procedure selects 50 records from a frozen deduplicated draft-search pool. The pool version, eligible sampling universe, random seed, selected IDs and reviewer decisions are retained. The project criterion is at least 80% raw agreement plus resolution of all conceptual disagreements; this threshold is not attributed to JBI. Cohen’s kappa is reported descriptively with prevalence limitations. If the criterion is unmet, the manual is clarified and a fresh pilot is performed. The 50-record pilot does not confer final study eligibility. Full-text boundary exercises follow before formal full-text screening.

Disagreements are discussed; unresolved decisions go to a predesignated adjudicator whose name must be supplied before formal screening. Changes to an eligibility interpretation are dated and applied to previously screened records when relevant. Operational records retain separate independent decisions and the consensus decision.

The screening manual defines this fixed primary exclusion hierarchy: FT01 ineligible source type; FT02 not original human research; FT03 ineligible population; FT04 no qualifying identifiable exercise context; FT05 no post-cessation sample; FT06 no baseline or suitable comparator; FT07 no eligible immune link; FT08 inseparable mixed intervention. Only the first established scientific reason is assigned at full-text exclusion; secondary concerns and supporting passages remain in notes. Unresolved information is not treated as an established failure. Duplicate reports/versions, not-retrieved reports and awaiting-classification records have administrative statuses outside the scientific exclusion hierarchy. The actual flow will reconcile identification, duplicates, screening, sought/retrieved reports, exclusions and included reports/cohorts, with the frontier register separate. Counts remain unavailable, not zero, until work is performed.

G. Data charting

G.1 Pilot, structure and verification

Ten purposively chosen reports will calibrate charting across exercise designs, conventional assays, cell/biofluid omics, early samples, repeated-bout monitoring and timing/population boundary cases. These reports are training material, not ten included studies, and may include cases ultimately excluded. Their source, selection rationale and unresolved questions are recorded in the pilot manifest. Pilot refinements are versioned before full extraction.

Two humans independently extract eligibility, timing, sample sizes, effects/uncertainty, precision domains, functional/clinical outcomes, validation and appraisal. One may enter purely bibliographic metadata for the other to verify. Discrepancies are resolved against source passages, with adjudication where necessary. The v2 dictionary, template and manual in `05_extraction` define the operational schema. Data use linked report, study and cohort tables with long-form assay/marker/time/comparison records. Raw independent extractions and reconciled data are retained separately.

G.2 Charting domains

- **Provenance:** Identifiers, publication/version status, correction/retraction checks and dates, study/cohort links, source locations, extractor and decision history.
- **Participants:** Eligibility and health definitions, infection screening, recruitment/analysis numbers, age, sex, training status, per-group/time/platform participant numbers and within-person pairing.
- **Exposure:** Design, exercise mode, duration, intensity metric, comparator, training background, previous load, co-interventions, environmental conditions and recovery schedule.
- **Sampling:** Exact time and time origin; matrix/immune compartment; collection, processing, storage, delay/freeze-thaw; circadian timing; plasma-volume/hydration correction; urinary dilution adjustment; salivary flow/secretion rate; cell composition and normalization.
- **Measurements:** Original and standardized marker names/identifiers, modality, targeted/untargeted assay, platform, units, detection/quantification limits, quality control, batch effects, molecular-identification confidence, bulk/single-cell structure and integration method.
- **Results:** Relevant contrasts, direction and magnitude, effect estimates and reported CI/SE or other uncertainty, multiplicity control, missing data, quality-control exclusions and measured/adjusted confounders. Relevant null and discordant findings are retained.
- **Precision domains:** All seven domains in C are charted separately: analytical reliability; temporal validity; immune specificity; individualization; functional/clinical linkage; independent validation and use; and feasibility. Subfields include personal baseline, within-person variation, test-retest/repeated-bout reliability, heterogeneity analyses, subgroup/individual model specification, validation type, participant-level separation, calibration/discrimination, thresholds and decision/implementation evidence where available.

- **Functions and outcomes:** Direct assays versus computed annotation; co-measurement versus tested association; outcome definition/ascertainment, symptom versus clinical versus pathogen outcome, biomarker timing relative to onset and follow-up duration.

For large omics datasets, chart the assay-level tested universe and all named immune candidates meeting the original aims/methods definition in accessible main text and supplements, regardless of significance. Retain reported feature counts and linked accessible supplements rather than inventing thousands of unreported rows. Record any candidate-selection rule and unavailable supplementary data. Preserve result units and avoid selecting only significant features. Unreported effect sizes are NR. Figure-derived extraction must identify the source figure/panel, reading or digitization method, software/settings, units and uncertainty handling; two reviewers independently check the source and values and retain their extractions and reconciliation. Estimated values are not presented as directly reported data. Where reliable extraction is impossible, use NR or UNCLEAR with the reason.

Participant count, biological sample count, cell count and technical repeats are different fields. The donor is the independent unit for single-cell inference; cells do not replace participants. Model splits are assessed at the participant/cohort level, including feature selection inside training folds. Raw individual change scores without measurement/within-person noise assessment do not establish reproducible responder status.

G.3 Time representation

Exact exercise-to-sampling times are primary. Optional display bins are: during exercise (ancillary); 0 to <30 minutes; 30 minutes to <3 hours; 3 to <24 hours; 24 to 72 hours inclusive; >72 hours; and NR. Bins are mutually exclusive display categories, not eligibility thresholds. Unknown timing is not imputed to a bin; uncertainty about exercise linkage may prevent main-map eligibility. Symptom/infection follow-up uses a separate time axis, with marker measurement before, concurrent with or after onset recorded explicitly.

H. Critical appraisal

Critical appraisal is prespecified to explain evidence limitations, not to create a clinical recommendation. Two reviewers independently apply tools matched to actual design, with discussion and adjudication. Current JBI tools are planned for randomized trials (2023 revision), quasi-experimental studies (2024), cohorts (2025), analytical cross-sectional studies (2026), and other eligible designs using the applicable current checklist [12]. Single-group pre/post studies are not treated as randomized trials. Crossover/repeated-measure limitations are charted explicitly. True diagnostic-accuracy questions use the current JBI diagnostic-accuracy tool. True individual-level future clinical-outcome or immune-response prediction-model development/validation studies additionally use PROBAST+AI (2025) [13]; a statistical model, molecular discovery classifier or simple association alone does not trigger that tool.

The exact official forms, versions, access dates and checksums will be archived before appraisal. Native response categories, domains and supporting source passages are retained. We do not assume all JBI forms have identical item structure or force them into a common score. The manual in `05_extraction/critical_appraisal_manual.md` governs design selection and overlap.

Assay/omics reporting and the seven precision domains are recorded separately from design appraisal. They include analytical reliability, pre-analytical handling, normalization, cell composition, repeated-measure analysis, multiplicity and independent validation. These custom descriptors are not a validated risk-of-bias tool. There is no quality sum, quality-threshold exclusion, GRADE certainty assessment or numerical readiness ranking. Appraisal findings contextualize the maps, including unresolved reporting limitations.

I. Synthesis and planned outputs

Synthesis is descriptive, stratified and narrative. Independent cohorts are the primary count for evidence density; report counts and distinct studies are also reported with explicit linkage. Platform features, cells, assays or time points are never used as proxies for evidence strength. Denominators are specific to each question and include applicable/NR/UNCLEAR information as appropriate. A/B overlap is disclosed, and the two stream totals are not simply added.

Three primary evidence maps are planned:

1. **Time by immune compartment/matrix:** Exact sampling coverage and optional time bins, exercise contexts and population strata.
2. **Candidate monitoring use by precision domain:** Separate domain evidence, assay reliability, individualization and practical implementation, with conventional and omics candidates distinguishable.
3. **Cohort-level validation links:** Technical confirmation, repeated bouts, independent participants/cohorts, directly measured function, tested associations and clinical/prediction outcomes, with explicitly labelled gaps.

The PRISMA flow is an additional process figure. Main tables describe study/cohort context, measurement/timing and limitations, individualization/validation evidence, and differences from adjacent reviews. Core A, support B and the frontier preprint register are visibly separated. Background sources do not enter original-study denominators. Counts of studies reporting a direction may be descriptive but are not a vote-counting estimate of effect or truth. Null, contradictory and unreported findings remain visible.

We will not pool effects, derive a diagnostic threshold, validate a marker, recommend return-to-training decisions or conclude that a candidate predicts infection from group differences alone. Discovery of a potentially homogeneous subset will be documented as a future focused-review opportunity. Any later decision to add an effect synthesis requires an explicit protocol amendment and appropriate methods before analysis. The manuscript will retain the scoping methodology regardless of the submission portal's article-category label [14].

J. Registration, governance and transparency

An OSF registration draft will be finalized and submitted by the authorized research team before formal five-database searching. No registration has been submitted during this design update. Public preliminary reconnaissance, citation verification and any strategy/calibration pilots preceding registration will be disclosed with dates, scope and their influence on decisions. The work is not described as wholly prospective from before any knowledge of the literature.

Substantive amendments record the previous rule, new rule, reason, date, work stage, information already known, authorizing team decision and need to revisit earlier records. Raw search exports and independent decisions remain immutable. The canonical protocol and execution appendices are versioned together.

This review uses published or lawfully accessible material and does not recruit participants or intervene in exercise. Institutional requirements for ethical review/exemption will be checked and recorded; no approval or exemption number is invented. Shareable search strategies, dictionaries, decision metadata and synthesis data are intended for an open repository; copyrighted full texts and restricted database exports are not redistributed contrary to their terms. Author names, affiliations, funding and competing interests are pending human declarations.

AI has supported public-source reconnaissance, evidence organization and drafting in this project, including coordinator and delegated Luna-agent work. Actual tools/models, dates, tasks, prompt/output references and human verification will be logged. AI is not an author,

human independent reviewer or adjudicator. Named humans remain responsible for eligibility, data, appraisal, citations and conclusions. No AI-produced citation or numerical claim is accepted without source verification. Final disclosure follows the then-current journal and institutional requirements.

K. Roles and execution plan

Roles to name are: principal investigator; reviewers A and B; search lead; independent PRESS reviewer; adjudicator; data/version manager; and analysis/writing lead. A second human reviewer and access to all five databases/full texts are available. Other roles may overlap where independence is preserved, but an adjudicator must be named before formal screening. Funding and competing interests cannot be inferred from the absence of information.

Use a reference manager with exportable identifiers and a tool that preserves blinded independent screening decisions; record software and versions before execution. The supplied JSON templates are specifications and exchange templates, not a claim that a production screening interface is already configured. The team may use its licensed screening software or controlled independent sheets with reconciliation; export and auditability are required.

Preparation is estimated at approximately 2–3 working weeks, depending on team availability: 3–5 working days to verify platforms and search strategies; 5–7 working days for reviewer training, pilot screening and charting, with parallel work where feasible; and 2–3 working days to resolve changes and finalize registration. These are planning estimates, not a promised completion date. Full-review effort will be estimated after the 50-record screening pilot and 10-report charting pilot, using actual yields and per-record effort. A provisional budget should distinguish two-person screening/extraction from single-person work and allow adjudication and update searches.

The local execution plan and registration draft specify stage gates and deliverables. Formal stages remain pending until the required human work is completed; completing this document does not certify PRESS review, calibration or registration.

L. Version history and decision status

Version 1.0 and associated build artifacts are preserved in the dated archive. Version 2.0 replaces the open-window-led title with candidate biomarkers for post-exercise precision monitoring; defines seven independent precision domains; separates acute A from strictly time-linked repeated-bout B; retains immediate samples without a 72-hour ceiling; expands searching to inception without a language filter; removes mandatory precision/biomarker/open-window terms; adds cohort linkage, complete outcome/validation fields, design-appropriate descriptive appraisal and AI transparency; and removes unsupported novelty claims.

The scope and operational design are ready for team calibration. Still pending are named administrative declarations, platform-specific verification/PRESS review, human pilots, archived appraisal forms, local software configuration and OSF submission. Subsequent formal searches, screening, extraction and synthesis are distinct execution stages and have not been performed.

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